

Day 2: Inspect the genomes

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Following Day 1, on this day, we're gonna learn how to analyse the genome assemblies to investigate the genomic features of the outbreak's isolates, such as their multi-locus sequence type (MLST), antimicrobial resistance genes, virulence factors, and other genomic features.

We will journey through the following topics:

1. **Assembly quality control:** How to assess the quality of the de novo assemblies.
2. **Species identification and MLST typing:** How to identify the species and strain of bacteria.
3. **AMR gene profiling:** How to identify antimicrobial resistance genes.
4. **Long-read sequencing & Hybrid assembly:** How to use long-read sequencing data to improve genome assembly.
5. **Genome annotation:** How to annotate the genome with functional information.
6. **Gene variation analysis:** How to identify gene variations between isolates.

Now, let's get started!

2.1 Analyse the assemblies

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Command-line for Post-Assembly Profiling Module

Post-Assembly Profiling builds on the genome assembly step by focusing on how to **evaluate the quality of assembled contigs** and **interpret their gene content**. In this section, you will learn how to assess assembly reliability and then explore what genes are present within the genome. The workflow includes two main steps:

1. Quality assessment of assemblies (e.g., completeness, contamination, N50, genome size)
2. Characterization of gene content (antimicrobial resistance (AMR) genes, virulence factors,...)

Command-line Workflow

From now on we will work mainly on Terminal (for MacOS / Linux user) or Command Prompt (for Windows user) or Visual Studio Code application.

Access to the class server

To begin, you need to access the class setup server. Open your Terminal and connect using the following command. This will log you into the server environment where all required tools and data for the session are prepared.

```
ssh -i microgen2026.pem user@136.112.198.45
-----
# microgen2026.pem: key identity file (private key for authentication)
# user: your server username
# 132.112.198.45: your server ip adress
```

Expore the materials available on the server

Go to your `day-2/post_assembly` directory using the following command:

```
cd /microgen2026/day-2/post-assembly/
```

List all files in the `post_assembly` directory:

```
ls
```

You should now see all the data needed for this module.

```
├─ post-assembly
│   ├── 925Kp_assembly.fasta
│   ├── 930Kp_assembly.fasta
│   ├── 937Kp_assembly.fasta
│   └─ ...
```

Count fasta files that we have for post-assembly module:

```
ls *.fasta | wc -l
-----
# ls: list files in your directory
# *.fasta: select all files ending with .fasta
# | : pipe sending output from left to right
# wc: word count tool
# -l: count number of lines
```

Question:

How many fasta files you have in your working environment?

Quality Assessment of Assemblies

For this section, **checkm2** is used

CheckM2 assesses the **quality of genome assemblies**. It estimates **completeness** (how much of the genome is recovered) and **contamination** (how much unwanted DNA is present) using a machine-learning approach based on gene content and genome features. This helps you determine whether your assembly is reliable before moving to downstream analysis.

Access conda environment

Activate the Conda environment containing the checkm2 tool:

```
conda activate microgen2026
-----
# conda activate: activate working enviroment
# microgen2026: our enviroment name contains all required tools for
               this workshop
```

Run checkm2 (completeness & contamination)

Check the availability of your tool by typing its name:

```
checkm2
```

The output should look like this:

```
1          ...::: CheckM2 v1.1.0 :::...
2
3  General usage:
4  predict      -> Predict the completeness and
    contamination of genome bins in a folder. Example usage:
5
6  checkm2 predict --threads 30 --input
    <folder_with_bins> --output-directory <output_folder>
7  testrun      -> Runs Checkm2 on internal test
    genomes to ensure it runs without errors. Example usage:
8
9  checkm2 testrun --threads 10
10 database
    -> Download and set up required CheckM2 DIAMOND database
    for annotation
11
12 Use checkm2 <command> -h for command-specific help.
```

Quality assessment of assemblies with checkm2:

```
checkm2 predict --threads 15 --input *.fasta --output-directory
    checkm2_output
-----
# checkm2 predict: run genome quality prediction
# --threads: number of CPU cores for parallel running, high cores count
    speed up the analysis
# --input: input your assembly fasta files
# --output-directory: location and file name to save results
```

After checkm2 finishes running, the output should look like this:

```
1  checkm2.log  diamond_output  protein_files
    quality_report.tsv
```

 **Question:**



How can you list the checkm2 output files in your working directory?

View checkm2 results

View the quality report using this following command:

```
1 cat checkm2_output/quality_report.tsv
2 -----
3 # cat: display the entire content of your file
```

The output should look like this:

```
1 Name      Completeness  Contamination
   Completeness_Model_Used Translation_Table_Used
   Coding_Density  Contig_N50  Average_Gene_Length
   Genome_Size GC_Content  Total_Coding-Sequences
   Total_Contigs  Max_Contig_Length  Additional_Notes
2 925Kp.assembly 100.0  0.19  Neural Network (Specific
   Model) 11 0.877  208194 308.6430039230338 5642876
   0.57 5353 134 670397 None
3 930Kp.assembly 100.0  0.2 Neural Network (Specific
   Model) 11 0.877  211821 308.6283846872082 5643929
   0.57 5355 141 670397 None
4 937Kp.assembly 100.0  0.19  Neural Network (Specific
   Model) 11 0.877  211826 308.5834889802017 5642843
   0.57 5354 131 670397 None
5 ...
```

Characterization of Gene Content

For this section, **Kleborate** is used

Kleborate characterizes **Klebsiella** and **Escherichia coli** genomes. It performs **MLST typing** and identifies important features such as **antimicrobial resistance (AMR) genes** and **virulence factors**. This allows you to understand the clinical and epidemiological significance of your isolates.

Access conda environment

Activate the Conda environment containing the kleborate tool:

```
conda activate microgen2026
-----
# conda activate: activate working enviroment
# microgen2026: our enviroment name contains all required tools for
  this workshop
```

If you are already in the conda environment microgen2026, please skip this step.

Run kleborate (version 3.1.3)

Check the availability of your tool by typing its name:

```
kleborate
```

The output should look like this:

```
1  usage: kleborate [-a ASSEMBLIES [ASSEMBLIES ...]] [-o
    OUTDIR] [-r] [--trim_headers] [--list_modules] [-p
    PRESET] [-m MODULES] [-h] [--help_all] [--version]
2
3  Kleborate: a tool for characterising virulence and
    resistance in pathogen assemblies
```

Check kleborate version:

```
kleborate --version
```

The output should look like this:

```
1  Kleborate v3.1.3
```

MLST typing and identification of AMR and Virulence genes using kleborate:

```
kleborate -p kpsc -o test_kleborate -a *.fasta -t 15
-----
# -p kpsc: specify the species complex, kpsc = klebsiella pneumoniae
    species complex
# -o: define the output file name
# -a: input your assembly fasta files
# -t: number of CPU cores for parallel running, high cores count speed
    up the analysis
```

After kleborate finishes running, the output should look like this:

```
1  klebsiella_pneumo_complex_output.txt
```

View kleborate results

View the kleborate output using this following command:

```
1 tail klebsiella_pneumo_complex_output.txt
2 -----
3 # tail: show the last part of your file
```

The output should look like this:

```

1
925Kp.assembly Klebsiella pneumoniae strong 134
208194 670397 5642876 no - ST147 3 4 6 1
7 4 38 78-1LV ybt 10; ICEKp4 3 12 4 11 3
21 18 3 10 16* 2 - 0 - - - - -
- - - - - - - - - - aadA2^;aph(3')-
VI;armA - - - - Mrx;mphA;mphE.v2^;msrE - -
sul1 tet(A).v1 - dfrA12 - - - -
NDM-1;OXA-48 SHV-11^ 35Q OmpK36GD -
GyrA-83Y;GyrA-87A;ParC-80I CTX-M-15?-0% - 2 7
12 wzl420 KL10 K10 Typeable ?10 98.94% - OL3a/
β 03αβ Typeable ?11 99.64% -

2
930Kp.assembly Klebsiella pneumoniae strong 141
211821 670397 5643929 no - ST147 3 4 6 1
7 4 38 78-1LV ybt 10; ICEKp4 3 12 4 11 3
21 18 3 10 16* 2 - 0 - - - - -
- - - - - - - - - -

3
4 ...

```

Output Interpretation

Download results to your local computer

Open another New Terminal window and run these commands on your New Terminal window.

Run this command to display your current directory (for MacOS/Linux users):

```
1  pwd
2  -----
3  # pwd: output your current directory
```

 For Windows users

```
1 echo %cd%
2 -----
3 # echo %cd%: output your current directory
```

The printout screen should look like this:

```
1 /Users/Downloads
```

💡 For Windows users

The printout screen should look like this:

```
1 C:\Users\Downloads
```

Download the `day-2/post-assembly/` folder from our learning server into your local computer:

```
1 scp -r -i microgen2026.pem user@136.112.198.45:~/
  microgen2026/day-2/post-assembly/ /Users/Downloads/
2 -----
3 # scp: secure copy (transfer files via SSH)
4 # -r: enable copy directories
5 # -i microgen2026.pem: key identity file (private key for
  authentication)
6 # user: your server username
7 # 132.112.198.45: your server ip adress
8 # ~/microgen2026/day-2/post-assembly/: result directory on
  learning server
9
  # /Users/Downloads/: destination on your local computer
  (as shown by "pwd" command)
```

💡 For Windows users

Simply replace `/Users/Downloads/` into `C:\Users\Downloads\`.

```
1 scp -r -i microgen2026.pem user@136.112.198.45:~/microgen2026/
  day-2/post-assembly/ C:\Users\Downloads\
```

Let's open the results and review them together

2.2 Long-Read sequencing & Hybrid assembly

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Hybrid Assembly with Long and Short Reads

For this section, **unicycler** is used

Unicycler is a genome assembler designed for bacteria that supports **short-read, long-read, and hybrid assembly**. It combines the accuracy of short reads with the long-range information of long reads to produce **high-quality, often complete genomes**, including circular chromosomes and plasmids

Access conda environment

Activate the Conda environment containing the unicycler tool:

```
conda activate microgen2026
-----
# conda activate: activate working enviroment
# microgen2026: our enviroment name contains all required tools for
  this workshop
```

Explore the materials available on the server

Go to your **day-2/hybrid_assembly** directory using the following command:

```
cd /microgen2026/day-2/hybrid-assembly/
```

List all files in the **hybrid_assembly** directory:

```
ls
```

You should now see all the data needed for this module.

```
|— hybrid-assembly
|   |— AMB201Kp_1.fastq.gz
```

```
| |— AMB201Kp_2.fastq.gz
| |— AMB201Kp_long.fastq.gz
| |— ...
```

Count fasta files that we have for post-assembly module:

```
ls *.fastq.gz | wc -l
-----
# ls: list files in your directory
# *.fastq.gz: select all files ending with .fastq.gz
# | : pipe sending output from left to right
# wc: word count tool
# -l: count number of lines
```

Question:

How many fastq files you have in your working environment?

Run unicycler (version 0.5.1)

Check the availability of your tool by typing its name:

```
unicycler
```

The output should look like this:

```
1
2      _
3     \|_\
4      \|_\
5      \|_\
6     \|_\
7     \|_\
8     \|_\
9     \|_\
10    \|_\
11    \|_\
12    \|_\
```

Hybrid assembly with unicycler:

```

unicycler -1 AMB201Kp_1.fastq.gz -2 AMB201Kp_2.fastq.gz -l
          AMB201Kp_long.fastq.gz -t 15 -o hybrid_AMB201Kp
-----
# unicycler: run genome unicycler for genome assembly
# -1: forward short reads from paired-end reads (read 1)
# -2: reverse short reads from paired-end reads (read 2) input: input
      your assembly fasta files
# -l: long read from nanopore sequencing
# -t: number of CPU cores for parallel running, high cores count speed
      up the analysis
# -o: location and file name to save results

```

After unicycler finishes running, the output should contain the following files:

```
1 ... assembly.fasta assembly.gfa unicycler.log
```

Question:

How can you list the unicycler output files in your working directory?

Long Read Assembly Only

You can perform **long-read-only** assembly using the following command. Ensure you **check read quality** before running the assembly:

```

1 unicycler -l your_long_reads.fastq -o longread_only -t 10
2 -----
3 # unicycler: run genome unicycler for genome assembly
4 # -l: your long read file from nanopore sequencing
5 # -t: number of CPU cores for parallel running, high cores count
      speed up the analysis
6 # -o: location and file name to save results

```

View unicycler results

View the assembly report using this following command:

```

1 cat hybrid_AMB201Kp/unicycler.log
2 -----
3 # cat: display the entire content of your file

```

The complete assembly should look like this:

```

1 Assembly complete (2025-09-14 14:09:25)
2 -----

```

```
3 Saving /Users/Downloads/hybrid_AMB201Kp/assembly.gfa
4 Saving /Users/Downloads/hybrid_AMB201Kp/assembly.fasta
```

Output Interpretation

Download results to your local computer

Open another New Terminal window and run these commands on your New Terminal window.

Run this command to display your current directory (for MacOS/Linux users):

```
1 pwd
2 -----
3 # pwd: output your current directory
```

For Windows users

```
1 echo %cd%
2 -----
3 # echo %cd%: output your current directory
```

The printout screen should look like this:

```
1 /Users/Downloads
```

For Windows users

The printout screen should look like this:

```
1 C:\Users\Downloads
```

Download the [day-2/hybrid-assembly/](#) folder from our learning server into your local computer:

```
1 scp -r -i microgen2026.pem user@136.112.198.45:~/
   microgen2026/day-2/hybrid-assembly/ /Users/Downloads/
2 -----
3 # scp: secure copy (transfer files via SSH)
4 # -r: enable copy directories
```

```
5 # -i microgen2026.pem: key identity file (private key for
   authentication)
6 # user: your server username
7 # 132.112.198.45: your server ip adress
8 # ~/microgen2026/day-2/hybrid-assembly/: result directory
   on learning server
9
   # /Users/Downloads/: destination on your local computer
   (as shown by "pwd" command)
```

For Windows users

Simply replace `/Users/Downloads/` into `C:\Users\Downloads\`.

```
1 scp -r -i microgen2026.pem user@136.112.198.45:~/microgen2026/
   day-2/hybrid-assembly/ C:\Users\Downloads\
```

Let's open the results and explore the assemblies with Bandage